

BLACK-BOX TESTING

The idea

Test what goes **in** and what comes **out**. Don't look at the code.

What you actually do

Make a table with 3 columns:

Class	Input	Expected Output
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One row for each *kind* of input. That's it.

What rows to write

Ask: what are all the different *kinds* of input?

Bad inputs (one row each):

- negative
- too big
- wrong type (3.14 when it wants a whole number)
- not a number ("ABC")
- empty / nothing

Good inputs (one row each):

- normal value
- any special value the task mentions (like 0)

Edges — if the task says a limit (e.g. max 12), add 3 rows: **11, 12, 13**.

Example: factorial(n)

Rules: no negatives, no decimals, no huge numbers. $0! = 1$.

Class	Input	Expected Output
Negative	-5	Error
Not whole number	3.14	Error
Not a number	"ABC"	Error
Too big	100	Error
Zero	0	1
Normal	7	5040
Edge below	11	works
Edge on	12	works
Edge above	13	Error

3 ways to lose points

1. Leaving out the Expected Output column.
2. Only testing good inputs (bad inputs are half the marks).
3. Testing 5 and 7 — same kind of input, so it's one row, not two.